Lecture 23 — From row operations to matrix products

Wed Oct 28

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Boyd video (optional): 0.0 0.0 1.0Lecture 27 — matrix multiplication · 0.0 0.0 1.0Lecture 26 — linear dynamical systems · 0.0 0.0 1.0ENGR108 playlist (Boyd)

Textbook: 0.0 0.0 1.0Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 10.1; review RREF and $x_{t+1} = Ax_t$ from 0.0 0.0 1.0Lecture 22. Related objectives: 0.0 0.0 1.0Lectures. 0.0 0.0 1.0Schedule.

Bring paper. Warm-up, fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, you should be able to:

1. Apply an elementary matrix to an augmented matrix and read the solution set (including a parametric form).
2. Classify unique / none / infinitely many from RREF and write infinite families with a parameter t.
3. Build a transition matrix from a two-compartment flow using a clear from/to convention, and compute one step $x_1 = Ax_0$.
4. Explain why two steps require $A(Ax_0)$ — a matrix–matrix idea — and compute a small product AB by hand.
5. Give an example with $AB \neq BA$, and compute A^2 and a small power $A^n x$.

Warm-up

You already multiply matrices when you left-multiply an augmented matrix by an elementary matrix. Let

$$[A \mid b] = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & 6 \end{array}\right], \qquad E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}.$$

(E is "add -2 times row 1 to row 2," built by doing that operation to I.)

(a) Compute $E\,[A \mid b]$.

(b) Read off the solution set of the resulting system. (It should be infinitely many solutions.)

Fill in

Solution types from RREF

After reducing $[A \mid b]$ to RREF, one of three things happens:

What you see in RREF	Conclusion	How to present the answer
A row $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$	_____ solution	Write "inconsistent" / empty set
Pivot in every variable column; consistent	_____ solution	Write the unique x
Consistent, and at least one free variable	_____ solutions	Set each free variable equal to a parameter (t, s, …) and express the pivot variables in terms of those parameters

Parametric form (one free variable). If x_2 is free, set $x_2 = t$ and solve for the others. The solution set is all vectors of the form

$$x = _____$$

as t runs over $\mathbb{R}$.

Transition matrices (from / to)

A discrete linear dynamical system is

$$x_{t+1} = Ax_t.$$

Convention used in this class (two compartments labeled $1, 2$):

- **Columns = "from."** Column j is the next state if the current state is e_j (everyone starts in compartment j).
- **Rows = "to."** Entry A_{ij} is the fraction (or amount) that goes **from** j **to** i.

So: read A_{ij} as "**to** i, **from** j."

One step is matrix–vector multiplication: $x_1 =$ _____. Two steps: $x_2 = Ax_1 = A(Ax_0)$. That is the same as $(AA)x_0$, so we need the **matrix–matrix** product $A^2 = AA$.

Matrix–matrix product

If A is $m \times n$ and B is $n \times p$, then AB is the $m \times p$ matrix whose jth column is

$$(AB)_j = _____.$$

Equivalently, entry (i, j) is the inner product of row i of A with column j of B.

Defined when? Number of _____ of A equals number of _____ of B.

Not commutative. In general $AB \neq BA$ (even when both are defined).

Powers. For a square matrix, $A^2 = AA$, $A^3 = A^2A = \cdots$, and $x_n = A^n x_0$.

Worked in class

W1. Unique solution. From the RREF

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array}\right],$$

write the solution.

W2. No solution. From the RREF

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 1 \end{array}\right],$$

explain why there is no solution.

W3. Infinitely many — parametrize. From the RREF

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right],$$

name the free variable, set it equal to t, and write $x = (x_1, x_2, x_3)$ in terms of t.

W4. Transition matrix. Two towns, Arcata and Eureka:

- Of those in Arcata, 80% stay and 20% move to Eureka.
- Of those in Eureka, 90% stay and 10% move to Arcata.

Order the state as $x_t = (\text{Arcata}, \text{Eureka})$. Using **rows = to**, **columns = from**, write A. With $x_0 = (5000, 7000)$, compute $x_1 = Ax_0$.

W5. One step vs two steps. Using the same A and x_0 as W4:

(a) Compute $x_2 = Ax_1$ (another matrix–vector product).

(b) Compute $A^2 = AA$ as a matrix–matrix product, then compute A^2x_0. Check that it matches x_2.

W6. Product by hand. Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix}.$$

Compute AB (columns Ab_1, Ab_2, or row–column dots).

W7. Order matters. With the same A and B, compute BA. Is $BA = AB$?

W8. Powers. Let $C = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

(a) Compute C^2 and C^3.

(b) Guess C^n, then check for $n = 2$ and $n = 3$. For $x_0 = (0, 1)$, compute $C^n x_0$.

Your turn

Y1. Let

$$[A \mid b] = \left[\begin{array}{cc|c} 1 & -1 & 2 \\ 3 & -3 & 6 \end{array}\right], \qquad E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}.$$

Compute $E\,[A \mid b]$ and write the solution set with parameter t.

Y2. From the RREF

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 5 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right],$$

write the general solution with free variable $t = x_3$.

Y3. Build A for two labs, North and South:

- From North: 70% stay, 30% go South.
- From South: 60% stay, 40% go North.

State order $x_t = (\text{North}, \text{South})$. Write A (rows = to, columns = from). If $x_0 = (100, 50)$, find x_1.

Y4. Let $A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$ and $x_0 = (1, 0)$. Compute $x_1 = Ax_0$ and $x_2 = Ax_1$. Then compute A^2 and A^2x_0.

Y5. With

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix},$$

compute AB and BA. Are they equal?

Answers (Your turn)

Y1. $E\,[A \mid b] = \left[\begin{array}{cc|c} 1 & -1 & 2 \\ 0 & 0 & 0 \end{array}\right]$. Free variable $x_2 = t$; then $x_1 - t = 2$, so $x_1 = 2 + t$. Solution: $x = (2 + t,\, t)$ for $t \in \mathbb{R}$.

Y2. Free $x_3 = t$. Then $x_1 - 2t = 5$ and $x_2 + 3t = -1$, so

$$x = \begin{bmatrix} 5 + 2t \\ -1 - 3t \\ t \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Y3. Rows = to, columns = from (order North, South):

$$A = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix}.$$

(Column 1: from North; column 2: from South.) Then

$$x_1 = Ax_0 = \begin{bmatrix} 0.7 \cdot 100 + 0.4 \cdot 50 \\ 0.3 \cdot 100 + 0.6 \cdot 50 \end{bmatrix} = \begin{bmatrix} 90 \\ 60 \end{bmatrix}.$$

Y4. $x_1 = (2, 1)$, $x_2 = Ax_1 = (4, 5)$. Also

$$A^2 = \begin{bmatrix} 4 & 0 \\ 5 & 9 \end{bmatrix}, \qquad A^2 x_0 = (4, 5),$$

same as x_2.

Y5.

$$AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Not equal: $AB \neq BA$.